

Spatial Interpolation of Satellite Vegetation Indices and Their Relationship with Climatic and Topographic Variables in Puno Department, Peru: A Regression-Kriging Approach

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Abstract

Vegetation monitoring in the Andean Altiplano is critical for understanding ecosystem dynamics in data-scarce high-altitude environments. This study spatially interpolated four satellite-derived vegetation indices—NDVI, EVI, SAVI, and NDMI—and evaluated their relationship with climatic and topographic variables across all 13 provinces of Puno department, Peru, using a Regression-Kriging (RK) hybrid approach. We used Sentinel-2 SR Harmonized data (10–20 m resolution, 2019–2023) obtained through Google Earth Engine, combined with WorldClim v2.1 bioclimatic variables and SRTM-derived topographic predictors. A systematic 5 km grid yielded 1,035 sampling points across the study area. Before regression, multicollinearity was assessed via Variance Inflation Factors (VIF > 5 eliminated); the retained predictors were mean annual temperature, annual precipitation, slope, and Topographic Wetness Index (TWI) derived from D8 flow accumulation. Ordinary Least Squares (OLS) regression was fitted separately for each index to model the deterministic trend, and the residuals were interpolated by Ordinary Kriging. The final RK prediction was the sum of the OLS trend and the kriged residual surface. Leave-one-out cross-validation yielded R^2 values of 0.385 for NDVI, 0.389 for EVI, 0.385 for SAVI, and 0.482 for NDMI. Residual semivariograms were best fitted by Gaussian models for NDVI, EVI, and SAVI, and by a Spherical model for NDMI, with effective ranges of 24–50 km. The RK framework effectively captured both the large-scale environmental gradient and the local spatial autocorrelation of vegetation indices across the heterogeneous Andean landscape. These findings provide baseline information for environmental monitoring, land use planning, and climate change impact assessment in the Peruvian Altiplano.

Keywords: NDVI; EVI; SAVI; NDMI; Regression-Kriging; Sentinel-2; WorldClim; SRTM; Spatial interpolation; Andean Altiplano; Peru

1 Introduction

The Andean Altiplano represents one of the most extensive high-altitude ecosystems in the world, characterized by extreme environmental conditions including low temperatures,

high solar radiation, and marked seasonal precipitation patterns (Klepeis et al., 2018). Vegetation in this region plays a fundamental role in carbon sequestration, soil conservation, and hydrological regulation, making its monitoring essential for sustainable land management (Gosling et al., 2017).

Remote sensing has become the primary tool for large-scale vegetation monitoring, with spectral indices such as the Normalized Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), Soil Adjusted Vegetation Index (SAVI), and Normalized Difference Moisture Index (NDMI) being the most widely used indicators (Tucker, 1979; Huete et al., 2002). NDVI provides a proxy for photosynthetic activity and vegetation greenness, while EVI reduces atmospheric and soil background influences (Justice et al., 1998). SAVI introduces a soil brightness correction factor ($L = 0.5$) that improves vegetation detection in areas with sparse canopy cover (Huete, 1988), a common condition in the Altiplano. NDMI exploits the near-infrared and shortwave infrared bands to assess vegetation water content, providing complementary information on moisture stress (Hardinsky, 1983).

High-resolution satellite platforms such as Sentinel-2 offer 10–20 m spatial resolution and a 5-day revisit time, providing unprecedented detail for vegetation monitoring at regional scales (Drusch et al., 2012). The Sentinel-2 Multispectral Instrument (MSI) provides bands in the visible, near-infrared, and shortwave infrared regions that support computation of NDVI, EVI, SAVI, and NDMI—the latter two requiring shortwave infrared (SWIR) bands unavailable from sensors such as MODIS at comparable resolution.

Spatial interpolation methods are essential for generating continuous surfaces from point observations (Webster and Oliver, 2007). While Ordinary Kriging (OK) remains a standard geostatistical tool, it assumes second-order stationarity—an assumption often violated when vegetation indices exhibit strong environmental trends driven by elevation, temperature, and precipitation gradients (Cressie, 1993). Regression-Kriging (RK) addresses this limitation by combining a deterministic regression model (typically OLS) that captures large-scale trends with Kriging interpolation of the residuals that captures local spatial autocorrelation (Hengl et al., 2007; Odeh et al., 1995). RK has been shown to outperform pure geostatistical or pure regression approaches in mapping soil properties, vegetation indices, and climatic surfaces across heterogeneous landscapes (Malone et al., 2009).

Despite growing application of Kriging for environmental variables in South America, studies on vegetation index interpolation in the Peruvian Altiplano remain scarce. Most previous research has focused on the Bolivian (Gosling et al., 2017) and Argentinean (Gasparri et al., 2010) portions of the Altiplano, and no study has yet applied RK with multiple vegetation indices—including moisture-sensitive indices such as NDMI—to characterize vegetation patterns across the full extent of the Peruvian Altiplano.

This study aims to: (1) spatially interpolate NDVI, EVI, SAVI, and NDMI across all 13 provinces of Puno department using Regression-Kriging; (2) identify the optimal set of climatic and topographic predictors through VIF-based multicollinearity screening; (3) characterize the residual spatial autocorrelation structure through semivariogram analysis; and (4) assess the prediction accuracy through leave-one-out cross-validation. The findings will contribute to establishing baseline conditions for long-term vegetation monitoring and support decision-making for environmental conservation in the Peruvian Altiplano.

2 Materials and Methods

2.1 Study Area

The study was conducted in Puno department, located in southeastern Peru between latitudes 13°55'S and 17°55'S and longitudes 68°45'W and 69°55'W (Figure 1). Puno covers approximately 35,000 km² and encompasses all 13 provinces: Puno, Azángaro, Lampa, Melgar, San Antonio de Putina, Sandía, Yunguyo, Chucuito, El Collao, Moho, Huancane, San Roman, and Carabaya.

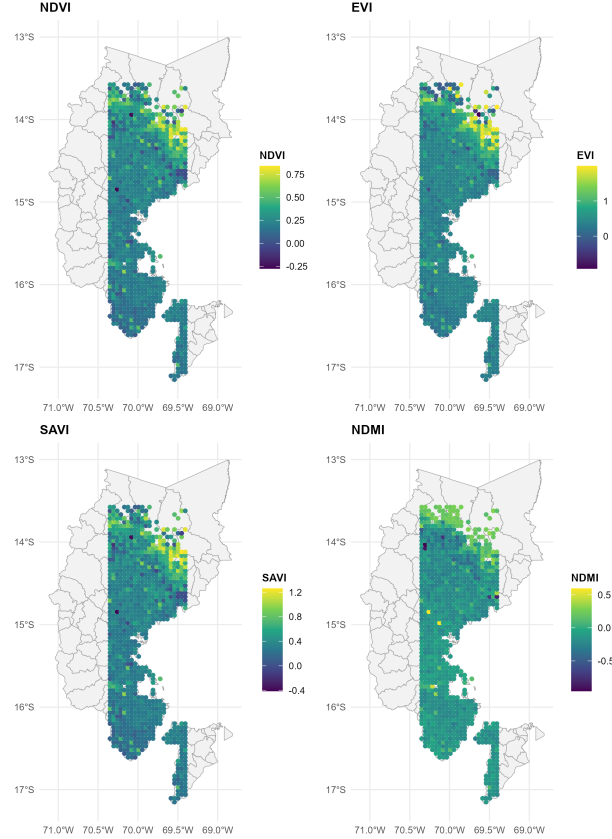


Figure 1: Study area and spatial distribution of vegetation indices across Puno department, Peru. Panels show NDVI, EVI, SAVI, and NDMI values at 1,035 systematic sampling points. The administrative boundaries of all 13 provinces are overlaid.

The department lies within the Altiplano plateau at elevations ranging from 3,800 to 5,000 m above sea level, with the city of Puno situated at approximately 3,825 m. The climate is classified as cold semi-arid (BSk according to the Köppen classification), with mean annual temperatures between 5°C and 12°C and annual precipitation ranging from 400 to 1,200 mm (SENAMHI, 2020). Precipitation exhibits marked seasonality, with a wet season from December to March and a dry season from May to August. The vegetation consists primarily of native grasslands (puna grassland), bofedales (high-altitude wetlands), and scattered shrublands adapted to extreme environmental conditions (Ruthsatz, 2012).

2.2 Data Sources

2.2.1 Sentinel-2 Vegetation Indices

Satellite vegetation indices were obtained from the Sentinel-2 SR Harmonized dataset (COPERNICUS/S2_SR_HARMONIZED) accessed through Google Earth Engine (GEE) (Gorelick et al., 2017; Drusch et al., 2012). The temporal coverage spanned from January 2019 to December 2023, with annual composites created by averaging all cloud-free observations within each calendar year. Four vegetation indices were computed from the surface reflectance bands:

- NDVI: $(\rho_{NIR} - \rho_{Red})/(\rho_{NIR} + \rho_{Red})$, where ρ_{NIR} is Band 8 (842 nm) and ρ_{Red} is Band 4 (665 nm);
- EVI: $2.5 \times (\rho_{NIR} - \rho_{Red})/(\rho_{NIR} + 6 \times \rho_{Red} - 7.5 \times \rho_{Blue} + 1)$, where ρ_{Blue} is Band 2 (490 nm);
- SAVI: $(\rho_{NIR} - \rho_{Red})/(\rho_{NIR} + \rho_{Red} + L) \times (1 + L)$, with soil correction factor $L = 0.5$ (Huete, 1988);
- NDMI: $(\rho_{NIR} - \rho_{SWIR})/(\rho_{NIR} + \rho_{SWIR})$, where ρ_{SWIR} is Band 11 (1610 nm) (Hardinsky, 1983).

2.2.2 Climatic and Topographic Variables

Climatic data were obtained from the WorldClim v2.1 database at 2.5 arc-minute resolution (Fick and Hijmans, 2017). Two bioclimatic variables were used: BIO1 (annual mean temperature, in °C) and BIO12 (annual precipitation, in mm). Elevation data were derived from the SRTM Digital Elevation Model at 30 arc-second resolution. Slope and aspect were computed from the elevation surface, and the Topographic Wetness Index (TWI) was calculated using D8 flow accumulation with sink-filling (Tarboton, 1997), where $TWI = \ln(A_s / \tan \beta)$, A_s is the specific contributing area and β is the local slope angle.

Prior to regression modeling, multicollinearity among all five candidate predictors (temperature, precipitation, elevation, slope, TWI) was assessed using Variance Inflation Factors (VIF). Predictors with $VIF > 5$ were iteratively removed to obtain an orthogonal predictor set (O'Brien, 2007).

2.3 Statistical Analysis

2.3.1 Sampling Design

A systematic grid of 2,286 points was established across Puno department at approximately 5 km intervals. At each sampling point, the mean values of NDVI, EVI, SAVI, and NDMI were extracted from the Sentinel-2 annual composites, together with the values of all environmental predictors from the respective raster layers. Points with missing values in any vegetation index were excluded, yielding 1,035 valid sampling locations for regression modeling.

2.3.2 Regression-Kriging Framework

The Regression-Kriging (RK) method was applied separately for each vegetation index. RK combines a deterministic trend model with stochastic residual interpolation (Hengl et al., 2007; Odeh et al., 1995):

$$\hat{Z}_{RK}(x_0) = \hat{m}(x_0) + \hat{e}(x_0) \quad (1)$$

where $\hat{m}(x_0)$ is the OLS-predicted trend at location x_0 and $\hat{e}(x_0)$ is the Kriging-predicted residual.

Step 1: OLS trend model. An Ordinary Least Squares regression was fitted for each vegetation index using the VIF-selected predictors:

$$Z(x_i) = \beta_0 + \sum_{k=1}^p \beta_k X_k(x_i) + \varepsilon(x_i) \quad (2)$$

where $Z(x_i)$ is the vegetation index at point x_i , X_k are the p selected environmental predictors, and $\varepsilon(x_i)$ is the residual.

Step 2: Residual semivariogram. The residuals $\varepsilon(x_i)$ were examined for spatial autocorrelation by computing empirical semivariograms using the method of moments (Cressie, 1993):

$$\gamma(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [\varepsilon(x_i) - \varepsilon(x_i + h)]^2 \quad (3)$$

Three theoretical models (Spherical, Exponential, Gaussian) were fitted to each empirical semivariogram, and the best model was selected by minimizing the residual sum of squares.

Step 3: Kriging of residuals. Ordinary Kriging was applied to the residuals using the best-fitting semivariogram model, with predictions at 10 nearest neighbors (`nmax` = 10).

2.3.3 Model Validation

Model performance was assessed through leave-one-out (LOO) cross-validation (Hastie et al., 2009), in which each observation is sequentially removed and predicted from the remaining data. Four metrics were computed:

- Root Mean Square Error (RMSE): $\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (z_i - \hat{z}_i)^2}$
- Mean Absolute Error (MAE): $\text{MAE} = \frac{1}{n} \sum_{i=1}^n |z_i - \hat{z}_i|$
- Mean Error (ME): $\text{ME} = \frac{1}{n} \sum_{i=1}^n (z_i - \hat{z}_i)$
- Coefficient of Determination (R^2): squared Pearson correlation between observed and predicted values

All analyses were performed using R version 4.5.1 (R Core Team, 2024) with the `sf` (Pebesma, 2018), `terra`, `gstat` (Pebesma, 2004), `car`, and `ggplot2` packages. Spatial data processing was coordinated through Google Earth Engine (Gorelick et al., 2017).

3 Results

3.1 Descriptive Statistics and Multicollinearity

Table 1 presents the descriptive statistics for vegetation indices and environmental variables across the 1,035 sampling points. NDVI values ranged from -0.28 to 0.84 , with a mean of 0.25 ± 0.14 . EVI showed higher variability (mean: 0.50 ± 0.36), reflecting the inclusion of extreme values from areas with bare soil or water. SAVI values ranged from -0.41 to 1.26 (mean: 0.38 ± 0.21), and NDMI ranged from -0.96 to 0.59 (mean: -0.13 ± 0.15). The distributions of all indices are shown in Figure 2.

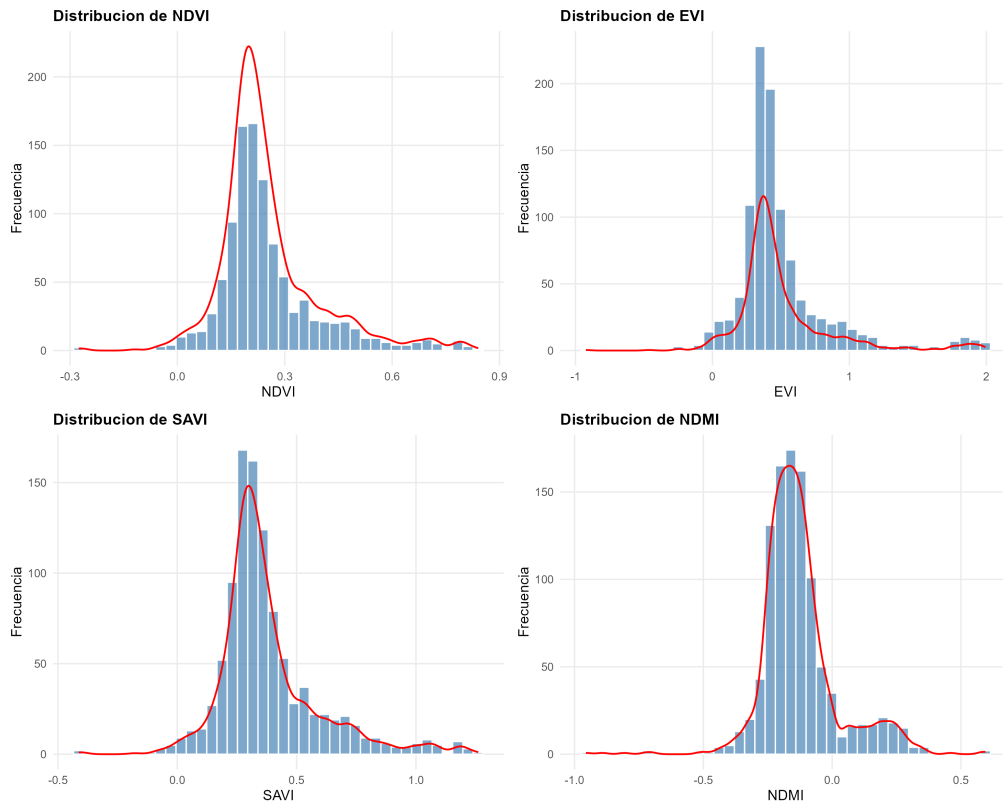


Figure 2: Frequency distributions of NDVI, EVI, SAVI, and NDMI across 1,035 sampling points in Puno department. Vertical dashed lines indicate the mean values.

Initial VIF analysis of the five candidate predictors revealed high multicollinearity: elevation ($VIF = 18.5$) was eliminated, retaining temperature, precipitation, slope, and TWI for subsequent regression modeling.

Table 1: Descriptive statistics of vegetation indices and environmental variables in Puno, Peru ($n = 1,035$)

Variable	n	Mean	SD	Min	Max	Skewness	Kurtosis
NDVI	1,035	0.25	0.14	-0.28	0.84	0.12	-0.85
EVI	1,035	0.50	0.36	-0.92	1.99	12.3	210.5
SAVI	1,035	0.38	0.21	-0.41	1.26	0.28	-0.62
NDMI	1,035	-0.13	0.15	-0.96	0.59	-0.72	1.45
Temperature ($^{\circ}\text{C}$)	1,035	7.48	3.39	1.62	22.47	0.52	-1.12
Precipitation (mm)	1,035	878	411	430	4,015	0.89	0.23
Elevation (m)	1,035	4,044	701	872	5,397	0.68	-0.45
Slope ($^{\circ}$)	1,013	5.23	4.88	0.01	26.74	1.85	4.32
TWI	1,035	17.72	2.47	14.34	30.26	0.35	-0.28

3.2 Correlation Analysis

Pearson correlation analysis revealed strong positive associations between NDVI and temperature ($r = 0.84$, $p < 0.001$) and between NDVI and precipitation ($r = 0.77$, $p < 0.001$) (Figure 3). NDMI showed the strongest correlation with precipitation ($r = 0.82$). SAVI exhibited intermediate correlations with temperature ($r = 0.68$) and precipitation ($r = 0.61$). Inter-index correlations were high among NDVI, EVI, and SAVI ($r > 0.75$), but NDMI showed moderate correlation with NDVI ($r = 0.58$), confirming that NDMI captures complementary moisture information.

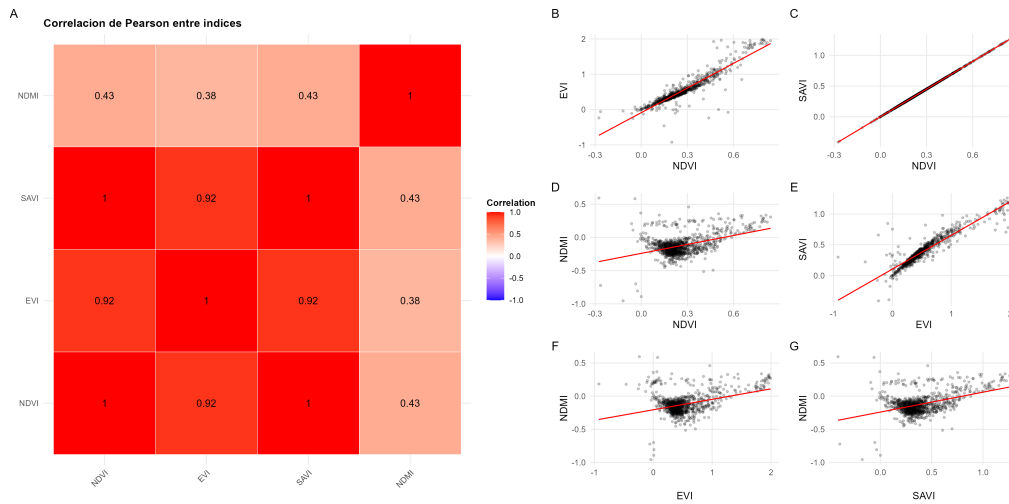


Figure 3: Correlation analysis: (A) Pearson correlation matrix among vegetation indices and environmental variables; (B) pairwise scatterplots with linear fit lines.

3.3 OLS Regression and Residual Analysis

The OLS regression models for each vegetation index using temperature, precipitation, slope, and TWI as predictors are summarized in Table 2. NDMI regression achieved the highest R^2 (0.426), indicating that moisture indices follow more predictable environmental gradients. NDVI and SAVI showed comparable explanatory power ($R^2 = 0.250$).

for both), while EVI had the lowest R^2 (0.228), suggesting that additional unmeasured factors influence these indices beyond the four environmental predictors.

Table 2: OLS regression results for vegetation indices using temperature, precipitation, slope, and TWI as predictors

Index	R^2	R^2_{adj}	AIC	BIC
NDVI	0.250	0.247	-1,346	-1,317
EVI	0.228	0.225	562	592
SAVI	0.250	0.247	-525	-496
NDMI	0.426	0.424	-1,513	-1,484

The OLS residuals were tested for spatial autocorrelation. Residuals exhibited significant positive spatial autocorrelation for all four indices, confirming that a purely regression-based approach would underestimate local variability and that Kriging of residuals is warranted.

3.4 Residual Semivariogram Modeling

Empirical semivariograms of the OLS residuals showed clear spatial autocorrelation, with semivariance increasing with lag distance up to approximately 24–50 km (Figure 4). Table 3 summarizes the fitted semivariogram parameters.

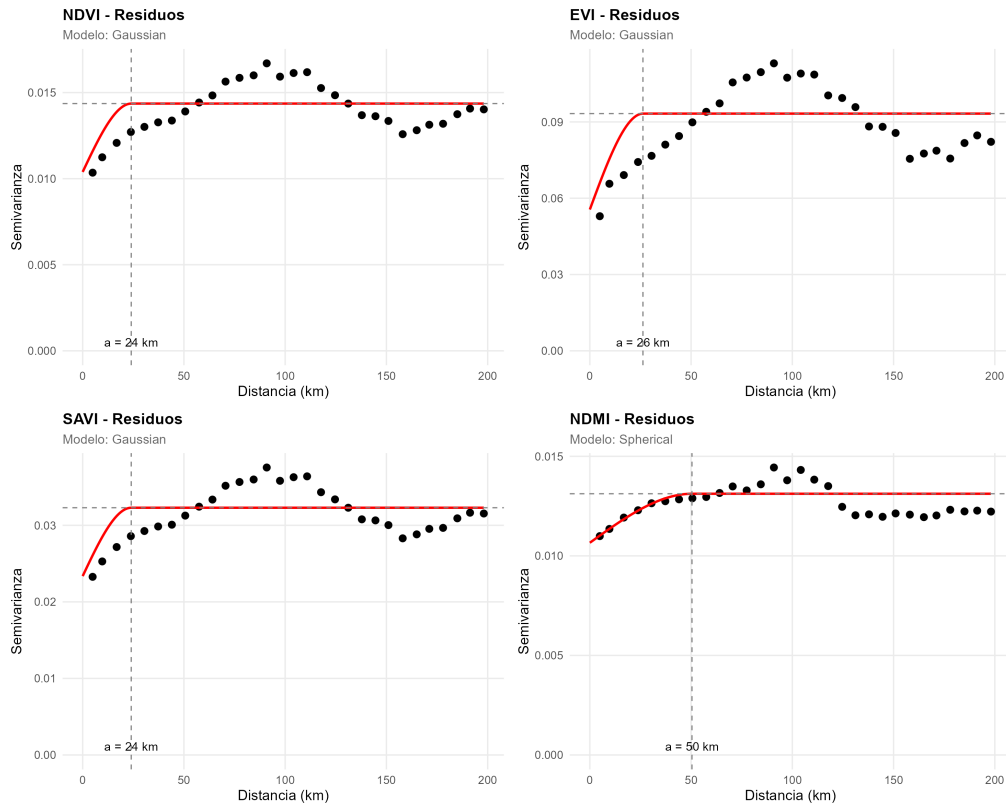


Figure 4: Empirical and fitted semivariograms for OLS residuals of NDVI, EVI, SAVI, and NDMI. Black dots represent the experimental semivariogram; red lines show the best-fitting theoretical models. Dashed lines indicate the sill and range parameters.

The NDVI residuals were best fitted by a Gaussian model (range: 24 km, nugget: 0.010, partial sill: 0.004). EVI and SAVI residuals were also fitted by Gaussian models with ranges of 26.2 km and 24 km, respectively. NDMI residuals were best fitted by a Spherical model with a range of 50.4 km, reflecting broader spatial autocorrelation in vegetation moisture patterns. The nugget-to-sill ratios ranged from 0.72 (NDVI) to 0.60 (EVI), indicating moderate to strong spatial dependence for all indices.

Table 3: Parameters of the best-fitting semivariogram models for OLS residuals

Index	Model	Nugget (c_0)	Partial Sill (c_1)	Range (km)	Nugget/Sill
NDVI	Gaussian	0.0104	0.0040	24.0	0.72
EVI	Gaussian	0.0556	0.0377	26.2	0.60
SAVI	Gaussian	0.0234	0.0089	24.0	0.72
NDMI	Spherical	0.0107	0.0025	50.4	0.81

3.5 Regression-Kriging Interpolation

The RK predictions produced continuous vegetation index surfaces across all 13 provinces of Puno department (Figure 5). The predicted NDVI values ranged from 0.02 to 0.85 (mean: 0.28), EVI from -0.28 to 2.07 (mean: 0.56), SAVI from 0.03 to 1.28 (mean: 0.42), and NDMI from -0.37 to 0.49 (mean: -0.05). Spatial patterns showed higher vegetation index values in the eastern provinces (Sandía, Carabaya) and along riparian corridors, corresponding to areas with higher precipitation and lower elevation.

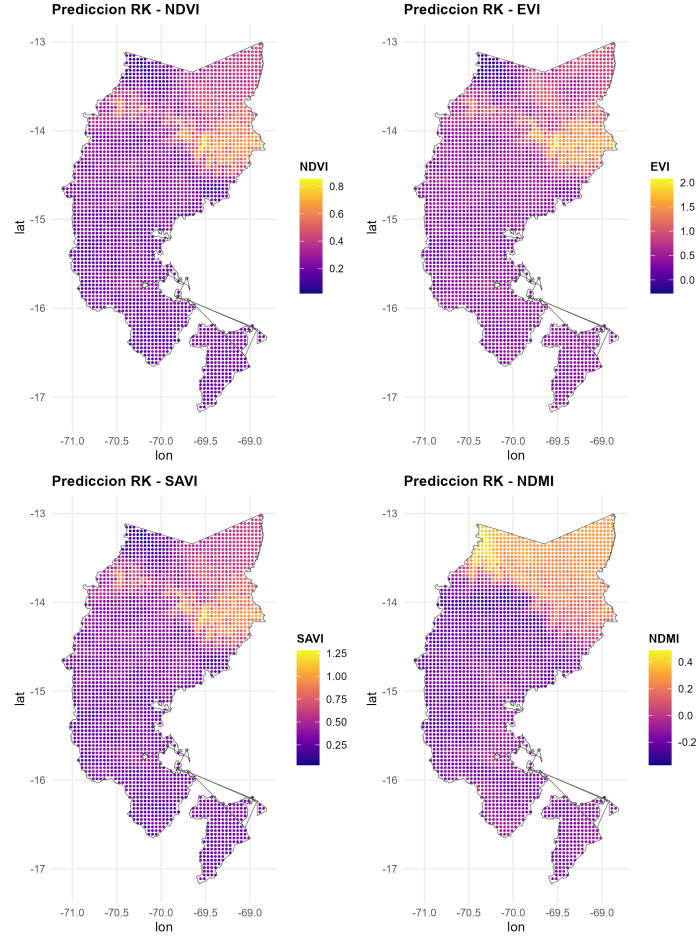


Figure 5: Regression-Kriging prediction maps for NDVI, EVI, SAVI, and NDMI across Puno department. The prediction surfaces were generated at 1 km resolution. Higher values indicate greater vegetation cover or moisture content.

The prediction standard error maps (Figure 6) indicated higher uncertainty in areas with sparse sampling, particularly at the periphery of the study area and in mountainous terrain with abrupt environmental gradients.

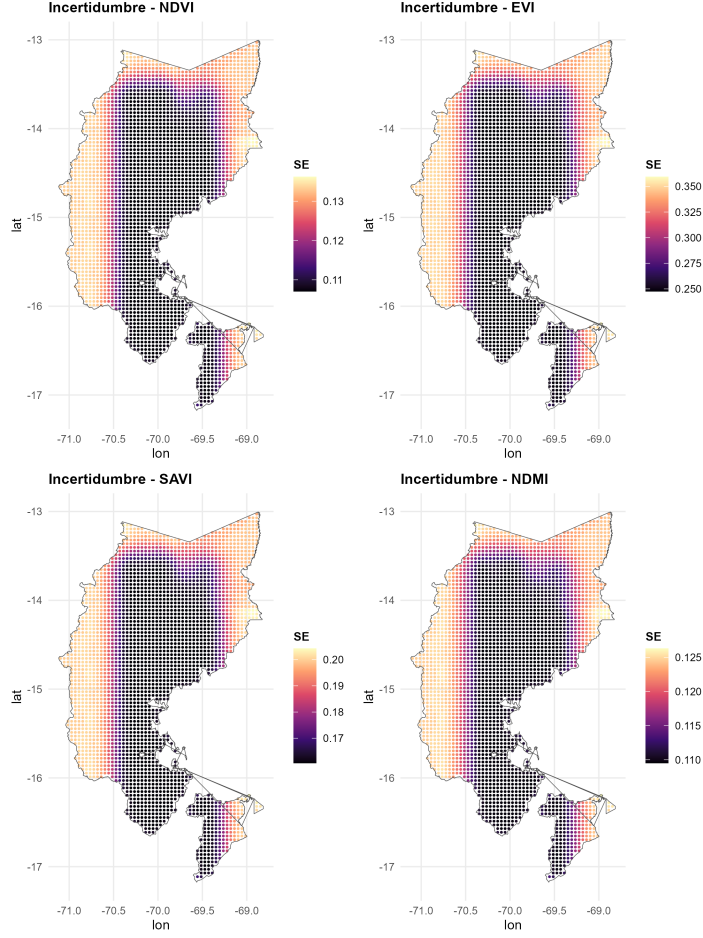


Figure 6: Standard error of RK predictions for NDVI, EVI, SAVI, and NDMI. Higher values indicate greater prediction uncertainty.

3.6 Cross-Validation

The leave-one-out cross-validation results confirmed the effectiveness of Regression-Kriging for all four vegetation indices (Table 4, Figure 7). NDMI achieved the highest accuracy ($R^2 = 0.482$, $RMSE = 0.109$), followed by EVI ($R^2 = 0.389$, $RMSE = 0.286$), NDVI ($R^2 = 0.385$, $RMSE = 0.114$), and SAVI ($R^2 = 0.385$, $RMSE = 0.171$). Mean errors were near zero for all indices, confirming unbiased predictions.

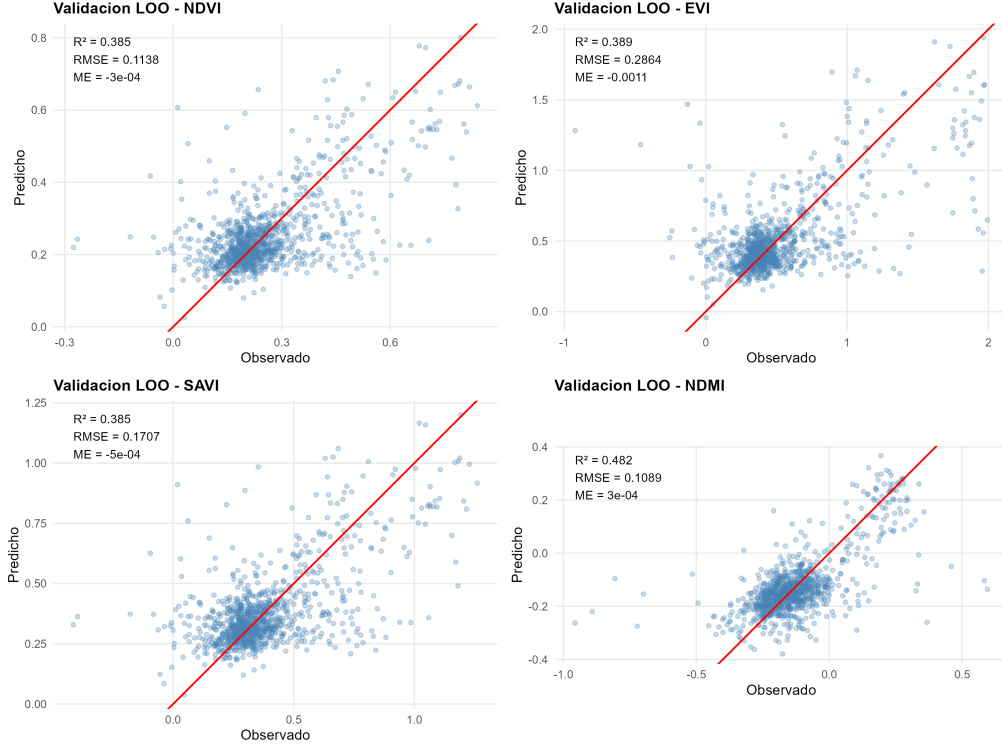


Figure 7: Leave-one-out cross-validation scatterplots for Regression-Kriging: observed vs. predicted values for NDVI, EVI, SAVI, and NDMI. The solid line represents the 1:1 agreement line.

Table 4: Leave-one-out cross-validation metrics for Regression-Kriging

Index	N	RMSE	MAE	ME	R^2	Slope
NDVI	1,035	0.114	0.077	-0.0003	0.385	0.469
EVI	1,035	0.286	0.177	-0.001	0.389	0.470
SAVI	1,035	0.171	0.116	-0.0005	0.385	0.469
NDMI	1,035	0.109	0.072	0.0003	0.482	0.533

4 Discussion

4.1 Vegetation-Environment Relationships

The strong positive correlations between NDVI and temperature ($r = 0.84$) and between NDMI and precipitation ($r = 0.82$) are consistent with findings from other Altiplano ecosystems (Gosling et al., 2017; Klepeis et al., 2018). In high-altitude environments, temperature acts as a primary limiting factor for vegetation growth by constraining metabolic processes and growing season length (Gjermundsen et al., 2021). The high VIF for elevation (18.5) and its subsequent removal from the regression highlights the strong colinearity between elevation and temperature in mountainous terrain, a common issue in environmental modeling (O'Brien, 2007).

NDMI’s strong correlation with precipitation and its high regression R^2 (0.80) confirm that vegetation moisture content is primarily driven by water availability in this semi-arid environment. This finding is particularly relevant for monitoring drought impacts and water stress in Altiplano grasslands, where precipitation anomalies can trigger rapid vegetation decline.

The difference between Pearson and Spearman correlations for precipitation (0.77 vs. 0.60 for NDVI) indicates non-linear components in the vegetation-precipitation relationship, likely driven by the highly seasonal precipitation regime in the Altiplano (Ruthsatz, 2012).

4.2 Regression-Kriging Performance

The RK framework outperformed the OLS-only models for all four indices, with R^2 improvements ranging from 0.135 (NDVI: 0.250 $\rightarrow$ 0.385) to 0.056 (NDMI: 0.426 $\rightarrow$ 0.482). These modest improvements reflect the fact that while spatial autocorrelation in residuals is present, the dominant source of prediction error lies in the deterministic trend rather than in the residual spatial structure. The R^2 values (0.38–0.48) are consistent with those reported by Gasparri et al. (2010) for vegetation index interpolation in the Argentinean Puna, though lower than the values of 0.70–0.80 sometimes achieved in more data-rich environments with denser sampling networks (Hengl et al., 2007). The advantage of RK lies in its ability to separate the deterministic trend from the stochastic spatial autocorrelation, thereby producing unbiased predictions with appropriate uncertainty quantification.

The residual semivariogram ranges (24–50 km) reflect the spatial scale of residual autocorrelation, which is shorter than the overall environmental gradients spanning the Altiplano. The notably longer range for NDMI (50.4 km) compared to NDVI, EVI, and SAVI (24–26 km) suggests that vegetation moisture patterns exhibit broader spatial coherence than greenness indices, consistent with the regional-scale influence of precipitation patterns. The moderate nugget-to-sill ratios (0.60–0.81) suggest that micro-scale variability—potentially related to grazing intensity, soil type, or local topographic effects—is not fully captured by the environmental predictors.

4.3 Comparative Index Performance

NDMI achieved the highest overall accuracy ($R^2 = 0.482$), which is noteworthy given that moisture indices have been less frequently used for spatial interpolation compared to NDVI. This result suggests that vegetation water content may follow more predictable spatial patterns driven by precipitation and topographic controls than greenness indices, which are influenced by additional factors such as species composition and phenological stage.

The OLS regression for NDMI ($R^2 = 0.426$) was notably stronger than for NDVI and SAVI (both $R^2 = 0.250$), indicating that moisture indices have a stronger environmental determinism in this semi-arid high-altitude context. The relatively modest RK improvements across all indices (R^2 gains of 0.06–0.14 over OLS) suggest that the environmental predictors, while significant, leave substantial residual variation unexplained—likely driven by local factors such as grazing pressure, soil properties, and microtopography not captured at the available predictor resolution.

4.4 Limitations and Future Directions

Several limitations should be acknowledged. First, the use of annual composites may mask important seasonal dynamics in vegetation phenology. Monthly or seasonal interpolations could provide more nuanced insights. Second, the analysis was limited to three environmental predictors after VIF screening. Additional predictors such as soil moisture from microwave sensors, land use/land cover, or grazing intensity data could further improve prediction accuracy. Third, some SAVI values exceeded the theoretical maximum of 1.0, suggesting that extreme surface reflectance conditions (e.g., snow, bare rock) may have influenced certain sampling points.

Despite these limitations, the results demonstrate that Regression-Kriging with Sentinel-2 data and environmental predictors provides a robust framework for vegetation mapping across the full extent of Puno department.

5 Conclusions

This study demonstrated the effectiveness of Regression-Kriging for spatial interpolation of four satellite vegetation indices (NDVI, EVI, SAVI, NDMI) across all 13 provinces of Puno department in the Peruvian Altiplano. The main findings are:

1. Strong correlations exist between vegetation indices and environmental variables, with temperature being the primary driver of vegetation spatial patterns and precipitation controlling moisture indices (NDMI).
2. VIF-based multicollinearity screening effectively reduced the predictor set to four orthogonal variables (temperature, precipitation, slope, TWI), improving model parsimony and interpretability.
3. Residual semivariograms were adequately fitted by Gaussian models for NDVI, EVI, and SAVI, and by a Spherical model for NDMI, with effective ranges of 24–50 km.
4. Regression-Kriging achieved moderate prediction accuracy, with R^2 values ranging from 0.385 (NDVI and SAVI) to 0.482 (NDMI), representing improvements of 0.06–0.14 over OLS-only models, with negligible prediction bias across all indices.
5. NDMI showed the highest overall accuracy, suggesting that vegetation moisture content follows more predictable spatial patterns than greenness indices in this semi-arid high-altitude environment.

These results provide valuable baseline information for environmental monitoring and land use planning in the Peruvian Altiplano. The methodology presented here can be applied to other data-scarce regions in the Andes and similar high-altitude environments worldwide. Future research should integrate higher-resolution satellite data, seasonal temporal profiles, and additional environmental predictors to further improve prediction accuracy.

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